GG-P-725B
July 14, 1978
SUPERSEDING
Int. Fed. Spec. GG-P-00725a (GSA-FSS)
June 30, 1964 and
Fed. Spec. GG-P-725
June 2, 1952

FEDERAL SPECIFICATION

PSYCHROMETERS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers psychrometers to be used for determining relative humidity of atmospheric air by means of wet- and dry-bulb temperature readings.

1.2 Classification.

1.2.1 Types. The psychrometers shall be of the following types, as specified (see 6.1):

Type I - Psychrometer, wall-mounted.
Type II - Psychrometer, sling.
Type III - Psychrometer, fan-ventilated, battery-operated.

Style A - Commercial.
Style B - Shipboard.

Type IV - Psychrometer, fan-ventilated, 115 volt, alternating current (a.c.) and direct current (d.c.).
Type V - Psychrometer, sling, pocket type.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple Wall.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

FSC 6685

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation, Methods of.
MIL-S-901 - Shock Tests, HI (High Impact); Shipboard Machinery, Equipment and Systems Requirements for.
MIL-L-10547 - Liners, Case, Waterproof.
MIL-E-16400 - Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore; General Specification for.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-810 - Environmental Test Methods.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 First article. Type III, Style B psychrometers furnished under this specification shall be products that have been tested and successfully passed the first article inspection specified herein. The first article shall be equivalent in all respects to the production units which the contractor proposes to furnish under this specification. Prior to submission of any production units for quality conformance inspection the contractor shall furnish two complete first articles for inspection and tests (see 4.3) to determine conformance with this specification. If the first article fails to meet this specification, the deficiencies should be corrected and the model resubmitted for approval. Approval of the first article by the command or agency concerned shall not relieve the contractor of this obligation to supply equipment conforming to this specification. Any deviations in production units from the approved first article shall be subject to the written approval of the command or agency concerned. All other type and style psychrometers covered by this specification, shall not be subject to first article inspection.

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3.2 Material and processes. The material used in the manufacture of the psychrometers shall be of the quality best suited for the purpose intended. The psychrometers shall be constructed of the lightest materials practicable consistent with ruggedness of design. Except as otherwise specified, Type III, Style B psychrometers shall be in accordance with MIL-E-16400 for exposed equipment.

3.2.1 Corrosion-resistant materials shall be used whenever practical. If corrodible materials are used they shall be protected to the fullest extent practicable from the effect of salt air or salt spray by plating, lacquer, baked enamel, or other covering or finish. Finishes which will crack, chip, or peel under normal usage or when subjected to changes of temperatures shall not be employed. When specified (see 6.1), metal parts shall be of brass, corrosion-resistant steel, or aluminum.

The use of aluminum in contact with nonferrous metals shall be avoided. All aluminum parts shall be anodized. All brass parts shall be nickel plated. Wear surfaces, pivots, bearing faces and similar items, which cannot be protected with finishes, shall be made of a material which will not appreciably tarnish or favor oxidation films.

3.2.2 The thermometer stem shall be made of standard glass thermometer tubing which shall withstand the temperature test specified in 4.5.3 to determine its suitability for the range of temperature to be measured (see 3.6.4). The bulb shall be of normal glass, clear, and of a quality that will not change appreciably with age.

3.3 Design and construction.

3.3.1 General requirements. The psychrometer shall consist of two identical nonmercurial thermometers, one fitted with a wick for wetting the bulb, thermometer holder or frame, and other components consistent with the type and style described herein. Relative humidity is determined from temperature readings of the wet and dry thermometers.

3.3.1.1 The psychrometer shall be so designed and constructed that motor heat and heat from hand holding the instrument will not influence the wet- and dry-bulb readings.

3.3.1.2 The basic difference of the five types of psychrometers covered herein shall be the means employed to convey air past the wick of the wet-bulb thermometer.

3.3.2 Thermometer. A thermometer shall consist of stem, expansion space, bulb, and indicator; and, in addition, for types I and II psychrometers, a terminal nib shall be provided (see figure 1). For Type V psychrometers, a button top shall be provided. When specified (see 6.1), the thermometer may be ordered separately without mountings.

3.3.2.1 Stem. The stem shall be cylindrical, with a round or oval capillary bore. It shall be provided with a strip of opalescent glass incorporated therein, running the full length of the stem, and subtending, at the rear, not less than 90 degrees of the circumference. The stem shall be 13/64 ± 1/64 inch in diameter.

3.3.2.2 Expansion space (excluding Type V). The expansion space shall be in the form of a tear drop as shown on figure 1. The design shall be such as to prevent breakage of the thermometer due to overheating. Each thermometer shall meet the test specified in 4.5.3.

3.3.2.2.1 Expansion space (type V only). The expansion space shall be a length of ungraduated capillary bore sufficient to provide over-temperature protection to 130°F when tested as specified in 4.5.3.1.

3.3.2.3 Bulb. The bulb shall be cylindrical in shape and shall be smoothly and neatly sealed to the lower end of the stem, and shall be concentric with it. The bulb shall be free of all foreign matter such as dust and chips of glass. It shall act as a reservoir for the thermometer liquid. Its walls shall be blown as thin as possible consistent with necessary strength in order to reduce the time lag of the thermometer to a minimum. The design of the bulb shall be such that the wick can be easily replaced without removing the wet-bulb thermometer from the back. The bulb shall not exceed 0.200 inch in diameter nor 7/8 inch in length (see figure 1).

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3.3.2.4 Indicator. Except when specified, the indicator for all psychrometers, other than type III, style B, shall be mercury. The substance of the mercury shall be clean and have a brilliant appearance. The indicator for type III, style B, psychrometers shall be an organic liquid and be black in color. There shall be no separation of the indicator in the thermometer tube because of improperly cleaned tube or expansion space. The space above the indicator shall be either evacuated or filled with an inert gas at the discretion of the contractor. The method of filling and the pressures employed shall conform to the best thermometer-filling practice.

3.3.2.4.1 When specified (see 6.1), the indicator shall be an organic liquid and shall be a red or black in color. The color stability characteristics of the liquid shall be tested as specified in 4.5.4.

3.3.2.4.2 Terminal nib (excluding type V). As a provision for holding the tubes in their proper relation to the thermometer frame, a nib approximately 3/32 inch long, drawn from the stem glass itself (not fused on), shall be formed at the upper end of the stem (see figure 1). It shall be neatly shaped and shall not prevent the tube from being held snugly against the mounting frame (see figures 2, 3, 4 and 5).

3.3.2.4.3 Button top (type V only). As a provision for holding the tubes in the frame of the pocket-type psychrometer, a button top approximately 3/64 inch larger diameter than the stem shall be formed from the stem glass itself at the upper end of the stem. It shall be neatly shaped and provided with a plastic sleeve to cushion the clip which holds it snugly against the frame (see figures 9, 10, and 11).

3.3.2.5 Thermometer range and graduation. The minimum acceptable length of the graduated scale shall be as shown on figure 1, unless otherwise specified (see 6.1). The scale graduations shall be etched in whole degrees on the stem, or the thermometer shall have permanent stain scales imprinted in the glass in whole degrees on the stem. The graduation lines shall be clear-cut, straight, and perpendicular to the axis of the stem. No appreciable variation in uniformity of spacing shall be noted at any point on the scale. When the thermometer is in a vertical position and viewed from the front, the minor graduation lines shall lie to the right of the axis and shall terminate in line with the axis, and the 5-degree and 10-degree lines shall extend across the axis to the left. The graduations shall be numbered at each multiple of 10 degrees. Those below zero shall be preceded by a minus sign (-). The figures shall be positioned to read with the thermometer horizontal, the bulb to the left; and shall be placed symmetrically about the line to which they refer. They shall be placed as near the axis as possible to facilitate readings. Graduation lines shall be not less than 0.005 nor more than 0.008 inch wide. The graduations shall terminate not closer than 1/2 inch from the top and bottom of the stem. Each thermometer tube shall bear the appropriate letter "F" or "C" near the top of the scale. See 3.3.2.5.1.

3.3.2.5.1 The thermometer range shall be one of the following as specified (see 6.1). Other ranges not to exceed a span of 150°F between minus 38°F and plus 135°F may be specified:

For types I and II:
- Plus 20° to plus 120°F.
- Minus 5° to plus 135°F.
- Minus 10° to plus 130°F.
- Minus 20° to plus 120°F.
- Minus 38° to plus 110°F.
- Minus 20° to plus 55°C.
- Minus 10° to plus 55°C.
- Minus 30° to plus 45°C.
- Minus 38° to plus 45°C.

For types III and IV:
- Plus 10° to plus 100°F.
- Plus 35° to plus 150°F.
- 0° to plus 55°C.
- Minus 20° to 40 plus 55°C.
- Minus 30° to plus 45°C.

For type V:
- Plus 30° to plus 110°F.

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3.3.2.6 Length. The total length of the bulb and stem shall be as shown on figure 1.

3.3.2.7 Accuracy. The maximum allowable error in thermometers shall be plus or minus 0.3°F for 0°F and above (plus or minus 0.2°C for minus 18°C and above), and plus or minus 0.5°F below 0°F (plus or minus 0.3°C below minus 18°C) when tested as specified in 4.5.2. The change in error for an interval of 20°F or 10°C shall be not greater than 0.5°F (0.3°C) on any part of the scale. The change in the ice point during a period of 90 days shall be not more than plus or minus 0.1°F (0.08°C). Each pair of thermometers furnished with a psychrometer shall be in agreement within 0.3°F (0.2°C) throughout their range.

3.3.2.8 Markings. When specified (see 6.1), each thermometer shall be marked near the upper end and to the right of the scale with the name or initials of the bureau or agency concerned and a serial number. The number shall be clearly visible when the tube is mounted in position on the frame.

3.3.2.9 Pigment. All etched graduations and numerals (see 3.3.2.5) shall be filled with a black pigment . The pigment shall not wash out or otherwise affect the readability of the thermometer when subjected to the tests specified in 4.5.14).

3.3.2.10 Stain. Permanent stain scales imprinted in the glass (see 3.3.2.5) shall be capable of passing durability of coloring material tests specified in 4.5.15.

3.3.2.11 Wick. The wick shall be of chemically clean muslin or bleached cotton sewn or woven into a tube. The wick shall be of such diameter that one end will fit snugly over the bulb of the wet-bulb (right-hand when facing thermometers) thermometer. The wick shall be treated such that absorption of water is readily achieved. For use on type I psychrometers, the wick shall be sufficient length to extend 3/8 to 1/2 inch beyond the bulb onto the stem and 1/4 to 3/8 inch below the bulb. For use on types II, IIIA and IIIB , and type IV psychrometer thermometers, the wick shall be sufficient to extend 1/2 inch beyond the bulb onto the stem and 1/2 inch below the bulb. Unless otherwise specified (see 6.1), a wick shall be securely fastened to the wet-bulb thermometer by means of a cotton thread. In addition, for type V psychrometer thermometers, the lower end of the wick shall be securely tied immediately below the wet-bulb thermometer by means of a cotton thread.

3.4 Type I, psychrometer, wall-mounted.

3.4.1 Corrosion-resistant metal back. Each thermometer shall be mounted upon a corrosion-resistant metal back. The design of the back shall be as shown on figure 5. The thermometer shall be mounted in such a way that it can be easily removed and readily remounted in the original position by the use of simple hand tools.

3.4.1.1 The stem of the thermometer shall be secured to the back by means of two U-shaped metal straps and machine-threaded screws. The straps shall be so formed as to fit the thermometer stem and hold it properly to the back.

3.4.2 Bracket. Each back (with thermometers mounted) shall be held in a position approximately 1-1/2 inches out from the mounting board by means of a corrosion-resistant metal bracket as shown on figure 6. Spacing rods or posts may be used in lieu of the bracket.

3.4.3 Mounting board. The mounting board shall be 11 inches long, 6 inches wide, and approximately 3/4 inch thick. The edges of the board shall be beveled. The wood used for the board shall be well seasoned, and shall be free from knots. The board shall be equipped with a hanger or hole for wall mounting.

3.4.3.1 Each metal back, with thermometer mounted, shall be secured to the bracket by means of two machine-threaded thumb nuts. Each bracket shall be attached to the mounting board, approximately 3 inches apart by means of two screws.

3.4.4 Water reservoir. A molded plastic or phenolic water reservoir with spout shall be provided for containing distilled water to wet the wick. The reservoir shall be approximately 1-1/2 inches in diameter and 2 inches high, and shall have a neat-fitting cover or cap. The cover or cap shall not have a hole in it. From the bottom edge of the reservoir, a spout 3/16 inch in diameter shall rise at an angle of 45 degrees to the height of the top of the reservoir. Distilled water shall not be provided in the water reservoir.

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considered to be the state of the art device like that installed on most aircraft. [illegible] accuracy is

3.4.4.1 The reservoir shall be mounted in a clamp holder at one end of a metal strip. The other end of this strip shall be permanently secured to the right-hand side of the mounting board. The strip shall be so placed that, when the reservoir is clamped in position, the upper end of the reservoir spout shall be directly under, and approximately 3/4 inch below, the right-hand thermometer bulb. The clamp holder shall be so constructed that there will be no interference with the wet-bulb thermometer when the reservoir is secured or removed.

3.4.4.2 When the reservoir has been mounted as specified in 3.4.4.1, the lower end of the wick shall be long enough so that it passes through the reservoir spout, and so that approximately 1 inch of its length rests inside, and on the bottom of, the reservoir. Therefore, with the upper end of the wick fastened to the bulb of the right-hand thermometer, and the other end inserted through the reservoir spout into the reservoir, the wick will lift the water from the reservoir by capillary action.

3.4.5 Shelter. When specified (see 6.1), a shelter shall be provided for the psychrometer.

3.4.5.1 The shelter shall be designed to house the psychrometer when installed for service use. The shelter shall be made of well-seasoned wood, shall be free from knots, shall be 3/4 inch thick; and shall be painted white. The inside dimensions of the shelter shall be: Depth, 5 inches; width, 8 inches; height in front, 16 inches; height at back, 18 inches. The back shall be solid, and shall extend lug-shaped beyond the top and bottom. The lug-shaped extensions shall be drilled at the end with a 1/4-inch screw hole for securing the shelter to the wall.

3.4.5.2 The shelter shall have a closed sloping top, a bottom pierced with 3/8-inch holes, louvered sides, and a louvered door. The door shall be fitted with two plated wrought steel hinges. The hinges shall be 1-1/2 inches long and 1-3/8 inches in width, when open. The door shall be fitted with a keylock. The length of the cylinder shall be such as to fit the thickness of the door.

3.4.5.3 The construction of the shelter shall be such that the enclosed psychrometer is exposed to the atmosphere on all sides, except the top and back; while the sloping top and louvered sides and door protect the psychrometer from rain, snow, and the direct rays of the sun.

3.4.6 Psychrometric table. Each type I psychrometer shall be provided with a psychrometric table showing, at normal barometric pressure, the percent relative humidity corresponding to either the wet- and dry-bulb thermometer readings or the dry-bulb thermometer reading and the difference between the readings. The psychrometric table shall be not less than 6 by 11 inches, and shall be legibly printed with permanent black ink. The printed table shall be unaffected by washing with soap and water. A calculator made of plastic may be furnished in lieu of the table. The table or calculator should be based on the psychrometric formula with its psychrometric constant adjusted for still air.

3.5 Type II, psychrometer, sling.

3.5.1 Thermometer frame. The thermometers shall be mounted parallel, 5/8 inch apart on centers; on a frame of corrosion-resistant metal fabricated as shown on figures 2 and 4. The metal used shall be in accordance with 3.2.1. The lower portions of the thermometers shall pass through guides formed by turning up the ends of the frame and drilling holes to receive the tubes (see figure 2). The two thermometers shall be so mounted that the air, after passing over the wet-bulb does not pass over the dry-bulb. The terminal nibs of the thermometers shall be located in the holes provided near the upper end of the frame. The upper ends of the thermometers shall be secured to the frame by means of two metal clips as shown on figure 7. A hole shall be drilled in the frame, 3/8 inch from the top and equidistant from the sides, and be reinforced with an eyelet to receive the swivel snap hook.

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3.5.1.1 The thermometer frame shall be straight before assembling so that, when the thermometers are mounted, no undue strain will be placed on them. The frame shall be smooth; and without burrs, or sharp or rough edges. Corners shall be slightly rounded so as to reduce the possibility of injury to the user.

3.5.2 Water reservoir. Unless otherwise specified (see 6.1), a water reservoir for containing distilled water shall be provided, but shall not be an integral part of the instrument. The reservoir shall be provided with a rubber stopper and special applicator for moistening the wick. Distilled water shall not be provided in the water reservoir.

3.5.3 Handle and sling. Unless otherwise specified (see 6.1), each type II psychrometer shall be provided with a comfortable sturdy handle, as shown on figure 8, made of hardwood such as birch, hard maple, or beech. A bearing shall be attached to the psychrometer end of the handle. To this bearing, two links and a neatly-shaped swivel snap hook, totaling a length of 3 inches, shall be secured so that the psychrometer may be attached for whirling. The snap hook shall fit, without binding, into the 3/16-inch hole provided in the thermometer frame. All metal parts shall be made of corrosion-resistant steel, nickel-plated spring steel, or nickel-plated brass.

3.5.3.1 There shall be no damage or distortion to the handle, sling, links or snap hook when tested as specified in 4.5.5.

3.5.4 Protective holder. When specified (see 6.1), a circumferential frame fabricated of corrosion resistive metal shall be furnished to protect the psychrometer from accidental breakage.

3.5.5 Instrument container. Unless otherwise specified (see 6.1), each psychrometer, including protective case, one pair of spare thermometers, 12 spare wicks, and a psychrometric table, shall be furnished in a wood case. Containers shall be made of any of the woods listed in group I of Federal Specification PPP-B-621; wood shall be seasoned in accordance with Federal Specification PPP-B-621. Wood shall be surfaced on both sides, and shall be of the following thickness: Bottom, sides and ends 1/4 ± 1/32 inch; top 3/16 ± 1/32 inch. Overall dimensions shall be determined by the manufacturer, and shall result in a finished box of such size as to accommodate the psychrometer and protective case when the handle is swiveled parallel and against the side of the case, without any undue arranging or crowding. The bottom of the box may be secured by either gluing or nailing. Where nailing is used, a minimum of three nails on each side and two on each end shall be used. When glue is used, a commercial grade waterproof glue shall be employed. Corners of the box shall be joined either by lock-corner or dovetail style joints; these shall be glued to provide a strong, tight joint. The cover of the box shall be made so as to slide in a groove or shall be hinged, having in this latter case a fastener to provide positive closure of the lid. The inner part of the box shall be provided with cushioned supports or clips to prevent movement or damage to the stowed equipment.

3.5.6 Psychrometric table. Unless otherwise specified (see 6.1), each type II psychrometer container (see 3.5.5) shall be provided with an easily-read psychrometric table giving, at normal barometric pressure, the percent relative humidity corresponding to either the wet- and dry-bulb thermometer readings or the dry-bulb thermometer reading and the difference between the readings. The table shall cover the range specified (see 6.1), and shall be based on the psychrometric formula with its constant adjusted for strong ventilation. A calculator made of plastic may be furnished in lieu of the table.

3.5.7 The psychrometric table shall fit into the instrument container and shall be at least 4 by 9 inches. It shall be made of a plastic material which is unaffected by washing with soap and water, and shall be lecibly printed with permanent waterproof black ink suitable for the application.

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3.6 Type III psychrometer, fan-ventilated battery operated.

3.6.1 General requirements. Type III psychrometers shall be a self-contained compact unit in a sturdy housing containing two identical thermometers (one fitted with a wick for wetting the bulb), thermometer holder or frame, intake shields, lamp for scale illumination and a fan coupled directly to a battery-powered motor. The overall dimensions of the psychrometer shall not exceed 10 by 4-1/2 by 2 inches, and the total weight shall not exceed 3-1/2 pounds.

3.6.1.1 Type III psychrometers, complete with thermometers, shall be designed to operate over the temperature range of the thermometers specified (see 3.3.2.5.1) and with relative humidities of 10 to 100 percent.

3.6.1.2 Type III psychrometers shall be so designed and constructed that motor heat, heat from the hand holding the instrument and other external sources of heat will not influence the wet- and dry-bulb readings.

3.6.1.3 The completed unit shall be sufficiently rugged to withstand normal shipment and use. The instrument shall be furnished complete, tested, and ready for service except as noted in 3.6.4.1 and 3.6.8.

3.6.1.4 Temperature. The type III, style B psychrometer only shall conform to the requirements of temperature range 2 of MIL-E-16400 except thermometer ranges shall be in accordance with 3.3.2.5.1.

3.6.1.5 Salt fog. The type III, style B psychrometer only shall conform to the salt fog requirements of MIL-E-16400.

3.6.1.6 Vibration. The type III, style B psychrometer only shall be capable of withstanding the type I vibration test of MIL-STD-167-1.

3.6.1.7 Shock. The type III, style B psychrometer only shall be capable of withstanding the grade A, class II, type A, lightweight shock test of MIL-S-901 as modified by 4.5.11.

3.6.1.8 Fungus. The type III, style B psychrometer only shall conform to the fungus requirements of MIL-E-16400.

3.6.1.9 Humidity. The type III, style B psychrometer only shall be designed to meet the humidity requirements of MIL-E-16400 except the instrument shall operate satisfactorily at any relative humidity from 10 to 100 percent over the temperature range specified in 3.3.3 and shall be capable of withstanding the test of 4.5.13.

3.6.2 Screws. All screws used in the psychrometer shall be class 2A thread.

3.6.3 Housing.

3.6.3.1 General requirements. The housing shall furnish support for the motor, thermometer frame, controls, lamp, intake shields, water bottle, and batteries. The housing shall be of a molded or fabricated construction which is rugged enough to permit satisfactory operation of the fan motor when the psychrometer is subjected to the stresses normally encountered for the use intended.

3.6.3.2 Battery compartment. The housing shall incorporate a compartment to accommodate the batteries of the power supply. The design of the compartment shall be such that the batteries can be easily replaced without disassembly of other parts of the psychrometer. Battery compartment, installation marking criteria and warning label for type III, style B psychrometers shall be in accordance with requirement 27 of MIL-STD-454. The warning label for type III, style B psychrometers shall also include marking to prohibit the use of mercury batteries.

3.6.4 Motor and fan. The motor and fan shall be direct-coupled and the combination shall be designed to draw a steady stream of air through the intake air duct and past the wet- and dry-bulbs of the thermometers. The design of the motor, fan and air duct shall be such that the air stream is directed across the thermometer bulbs before reaching the motor.

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3.6.4.1 Motor. The motor shall be a d.c. type designed to operate from not more than three size D dry cell batteries. Batteries shall not be furnished with the psychrometer. The motor shall be sturdy in construction, requiring no lubrication in service. It shall be a standard, readily available commercial design.

3.6.4.2 Motor-mount. The motor shall be mounted in the psychrometer housing in such a manner that, in event of motor failure, the motor and fan assembly can be easily removed and replaced as a unit without the use of special tools.

3.6.4.3 Motor controls. A positive means for starting and stopping the motor shall be provided. Controls shall also be provided to permit independent operation of the motor and the scale illumination lamp or lamps (see 3.6.6).

3.6.4.4 Fan. The fan shall draw air across the thermometer bulbs at a velocity of not less than 15 feet per second (see 4.5.6).

3.6.5 Air duct. The air duct shall be designed to give optimum air flow at the thermometer bulbs and shall incorporate intake thermal shields for the thermometers. Since the heat of the observer's body can affect psychrometer observations in cold weather by heating the intake air, the design of the air duct shall be such as to inhibit the drawing of air from the vicinity of the observer's face or body when observations are being made. The intake thermal shield shall be as thin as is consistent with sturdiness and ability to withstand hard usage, and shall be of suitable material and be finished with a reflective finish to minimize the influences of instrumental temperature upon the thermometer bulbs. The shield shall be of such design that the wick on the wet-bulb thermometer can be wetted and replaced with ease.

3.6.6 Illumination. The psychrometer shall incorporate an illumination system to provide for illumination of the thermometer scales. The system shall consist essentially of one or more lamps mounted in such a manner that the scales of the thermometers are totally and evenly illuminated. The lamp or lamps employed shall be of a standard, readily available type, and shall be designed to operate from the same power supply as the motor. It shall be possible to replace the lamps with a minimum of disassembly of parts. Illumination provided by the system shall be of the indirect type and shall be red in color, to provide for adequate protection of the "dark conditioning" of the eye. Means shall be provided for varying the intensity of the scale illumination.

3.6.7 Thermometer mount. A mount for holding the wet- and dry-bulb thermometers shall be provided to fit the thermometers specified in 3.3.2. The thermometers shall be recessed in the mount to reduce the chance of accidental breakage during handling. The thermometers shall have rubber sleeves at each end, fitted into the mount recess in order to protect the glass tubes against breakage from shock. The mount shall position the thermometers on the psychrometer so that the graduations and numerals of both thermometers will be clearly visible from a point in front of the instrument. The mount shall not interfere with the wick specified in 3.3.9 and shall position the thermometer so that the 1/2 inch wick extension is always within the psychrometer housing. The design of the mount shall be such that, while the ventilation takes place, it shall be possible to observe the thermometer scales at all times without interruption in the operation of the instrument, so that observation of the true maximum wet-bulb depression will not be missed. The mount shall be so constructed that the thermometers can be removed for purposes of replacement with a minimum of disassembly of parts and without the use of special tools.

3.6.8 Water bottle. The water bottle shall be made of a plastic material, polyethylene or similar, and shall be a minimum 1-ounce capacity. The bottle shall provide a means of applying distilled water to the wick of the wet-bulb thermometer. Means shall be provided in the housing for holding the minimum 1-ounce water bottle when not in use. Distilled water shall not be provided in the water bottle.

3.6.9 Identification plate. An identification marking of a size consistent with legibility and space available shall be securely fastened on the exterior of the psychrometer housing. The identification for style B psychrometers shall be a plate, except adhesive backed plates shall not be used, and shall include the following information.

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Bureau or agency----------

Instrument name or model No.----------

Serial No.----------

Name of manufacturer----------

Contract No.----------

Stock No.----------

U.S.----------

Stock number and serial numbers will be supplied by the procuring activity. The above required information may be printed or stamped on the identification label. Commercial type labels are acceptable for style A psychrometers.

3.6.10 Carrying case. A sturdy metal or plastic carrying case, properly padded to guard the instrument against shock and vibration, shall be provided. The case shall contain a woven nylon strap with a snap-on clip capable of being adjusted in length so as to serve as a handle or shoulder strap. The finish of the case shall be the same as that of the psychrometer housing. The carrying case for style B psychrometers shall contain provision for storing the repair parts specified in 3.7.

3.6.10.1 Carrying case components. The carrying case shall contain an extra water bottle of a minimum 1-ounce capacity with a means of applying distilled water to the wick of the wet bulb, a technical manual, a small container of proper size wicks with cotton thread (see 3.3.2.11), and a plastic calculator for determining dew point and relative humidity from the psychrometer readings. The calculator shall be based on the psychrometric formula with its psychrometric constant adjusted for strong ventilation and may be by slide rule, tables, or graphs. The calculator shall not introduce an error greater than that introduced by a 0.2 degree difference error in the thermometer readings at any point in the range of the psychrometer. When specified for style A psychrometers and mandatory for style B psychrometers, the carrying case shall contain provision for storing the repair parts specified in 3.7.

3.6.11 Mounting bracket. When specified (see 6.1), a bracket designed for wall mounting of the psychrometer shall be furnished. The bracket shall provide for mounting of the psychrometer in the normal reading position and with the air duct farthermost from the wall. The means of attaching the psychrometer to the bracket shall be such that the psychrometer may be readily removed for portable use without the use of special tools.

3.7 Repair parts. When specified (see 6.1), the following repair parts in quantities as specified by the procuring activity shall be furnished with each psychrometer:

(a) Thermometers.
(b) Water bottles.
(c) Illumination lamps.
(d) Wicking.
(e) Screws (for housing and thermometer mount).

3.8 Technical manuals for type III, style B psychrometers only shall be prepared in accordance with the data ordering documents specified (see 6.2).

3.9 Type IV psychrometers, fan-ventilated (115 volt, a.c. and d.c.).

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3.9.1 General requirements. The type IV psychrometer shall be similar in all respects to the type III psychrometer specified in 3.6, except that references to d.c. motor, batteries, and those specific to style B psychrometers do not apply.

3.9.2 Motor. The motor shall be of the universal type designed to operate on 110-120 volts, 60 cycles and direct current. The motor shall be of sturdy construction, and shall require no lubrication in service. It shall be a standard, readily available commercial model.

3.9.3 Power cord and plug. The motor leads shall be connected to type V, 2 conductor size 16 cord. Cord length outside of the motor housing shall be at least 6 feet. A three prong plug shall be attached to the free end of the cord. The cord shall be supported inside of the motor housing in such a manner that no mechanical stress can be transmitted directly to the electrical connections. Where the cord enters the motor housing, the passage hole shall be provided with a rubber grommet to prevent fraying of the cord.

3.10 Type V, psychrometer, sling, pocket-type.

3.10.1 Thermometer frame. The thermometers shall be mounted parallel, 5/8 inch apart on centers; on a frame of corrosion-resistant metal fabricated as shown on figures 9 and 10. The metal used shall be in accordance with 3.2. The lower portions of the thermometers shall pass through guides formed by turning up the ends of the frame and drilling holes to receive the tubes (see figure 10). The two thermometers shall be so mounted that the air, after passing over the wet-bulb does not pass over dry-bulb. The button heads of the thermometers shall be located in the punch outs provided near the upper end of the frame. The upper ends of the thermometers shall be secured to the frame by means of two metal clips as shown on figure 11. The clips shall be fastened to the frame by means of two screws and two lock washers as shown on figure 10. A hole shall be drilled in the frame, 3/8-inch from the top and equidistant from the sides.

3.10.1.1 The thermometer frame shall be straight before assembling so that, when the thermometers are mounted, no undue strain will be placed on them. The frame shall be smooth; and without burrs, or sharp or rough edges. Corners shall be slightly rounded so as to reduce the possibility of injury to the user.

3.10.2 Water-reservoir. When specified (see 6.1), a water reservoir for containing distilled water shall be provided, but shall not be an integral part of the instrument. The reservoir shall be provided with a rubber stopper and special applicator for moistening the wick. Distilled water shall not be provided in the water reservoir.

3.10.3 Handle and sling. Each type V psychrometer shall be provided with a 3-3/4 inches long aluminum handle, partly knurled, as shown on figures 12 and 13. The metal used shall be aluminum rod 5/16 inches in diameter. The handle shall be equipped with a 4-inch length of #12 American pattern nickel-plated brass chain with each end having a link consisting of two complete loops. One link shall be inserted through the hole in the frame and the other link shall be attached to the ring as shown on figure 10. The ring shall be a split-ring and shall be made of a hard steel.

3.10.3.1 There shall be no damage to the handle and sling, when tested as specified in 4.5.5.2.

3.10.4 Pocket case. Each psychrometer shall be furnished with a pocket case, fitted with a protective aluminum insert (see figure 14), and designed to contain and protect the complete instrument, including the handle and chain.

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3.10.5 Instrument container. Unless otherwise specified (see 6.1), each psychrometer, including pocket case, one pair of spare thermometers, 12 spare wicks, and a plastic calculator, shall be furnished in a wood case. Containers shall be made of wood and shall be seasoned in accordance with Federal Specification PPP-B-621. Wood shall be surfaced on both sides, and shall be of the following thickness: Bottom, sides and ends 1/4 ± 1/32 inch; top 3/16 ± 1/32 inch. Overall dimensions shall be determined by the manufacturer, and shall result in a finished box of such size as to accommodate the psychrometer, enclosed in the pocket case, the spare thermometers, the spare wicks, and the calculator, without any undue arranging or crowding. The bottom of the box may be secured by either gluing or nailing. Where nailing is used, a minimum of three nails on each side and two on each end shall be used. When glue is used, a commercial grade waterproof glue shall be employed. Corners of the box shall be joined either by lock-corner or dovetail style joints; these shall be glued to provide a strong, tight joint. The cover of the box shall be made so as to slide in a groove or shall be hinged, having in this latter case a fastener to provide positive closure of the lid. The inner part of the box shall be provided with cushioned supports or clips to prevent movement or damage to the stowed equipment. In lieu of a wood case, a molded plastic case shall be considered acceptable.

3.10.6 Calculator. Each type V psychrometer shall be provided with a plastic calculator.

3.11 Workmanship. Each psychrometer shall be free from defects which detract from its appearance or which may impair its serviceability. The finished psychrometer shall perform its functions in a manner suitable for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection (type III, style B only). First article inspection shall be conducted on a psychrometers and shall consist of the tests in table I. Shock and vibration tests shall be conducted before completion of other tests and the fungus test shall be conducted after completion of other tests except the pigment stability test.

TABLE I. First Article Tests for Type III, Style B, Psychrometers Only

Test	Requirement paragraph	Test paragraph
Surface examination		4.5.1
Thermometer accuracy	3.3.2.7	4.5.2
Expansion space	3.3.2.2	4.5.3
Color stability	3.3.2.4	4.5.4
Air velocity	3.6.4.4	4.5.6
Low temperature	3.2.2 and 3.6.1.4	4.5.7
High temperature	3.2.1, 3.3.2.2 and 3.6.1.4	4.5.8
Salt fog	3.6.1.5	4.5.9
Vibration	3.6.1.6	4.5.10
Shock	3.6.1.7	4.5.11
Fungus	3.6.1.8	4.5.12
Humidity	3.6.1.9	4.5.13
Pigment stability	3.3.2.9	4.5.14
Stain durability	3.3.2.10	4.5.15

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4.3.1 First article test report (type III, style B only). A first article test report shall be furnished in accordance with 6.3.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of group A, B and C tests in accordance with 4.4.3, 4.4.4 and 4.4.5. Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.4.1 End item inspection. The sample unit for this inspection shall be one complete psychrometer.

4.4.2 Rejected lots. If any inspection lot is rejected, the supplier may withdraw the lot and rework it to correct defects, or screen out defective units, as applicable, and reinspect. Such lots shall be separated from new lots and shall be clearly identified as reinspected lots. Rejected lots shall be inspected using tightened inspection in accordance with MIL-STD-105.

4.4.3 Group A tests. A random sample of psychrometers shall be subjected to Group A tests shown in table II. Sampling inspection shall be based on MIL-STD-105. The inspection level shall be level I with an acceptable quality level (AQL) of 1.5 for major defects and 4.0 for total defects. Any psychrometer or case in the sample containing one of more visual or dimensional defects or failing to comply with any of the tests shall be rejected, and if the number of defective psychrometers or cases in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. All type III, style B psychrometers offered for delivery shall be subjected to group A tests. Any type III, style B psychrometers failing any group A test shall be rejected.

4.4.4 Group B tests. A random sample of psychrometers shall be selected from each lot in accordance with inspection level S-2 of MIL-STD-105 and shall be subjected to the group B tests of table II. The acceptable quality level (AQL) for test failures shall be 1.0 percent. If any type III, style B psychrometer fails any group B test the entire lot of type III, style B psychrometers shall be rejected.

4.4.5 Group C tests (type III, style B only). When required by the Government (see 6.1) 2 psychrometers shall be selected at random and subjected to the group C tests in table II. Shock and vibration tests shall be conducted prior to other tests. Failure of any one of the tests shall be cause for rejection of the entire lot represented by the sample.

TABLE II. Quality conformance tests

Test	Requirement paragraph	Test paragraph
GROUP A (all types)		
Surface examination		4.5.1
GROUP B (all types)		
Thermometer accuracy	3.3.2.7	4.5.2
Expansion space (excluding type V)	3.3.2.2	4.5.3
Color stability	3.3.2.4	4.5.4
Handles and slings (types II and V)	3.5.2	4.5.5
Air velocity (types III and IV)	3.6.4.4	4.5.6
Expansion space (type V)	3.3.2.2.1	4.5.3.1
Pigment stability	3.3.2.9	4.5.14
Stain durability	3.3.2.5	4.5.15
GROUP C (type III, style B)		
Low temperature	3.2.1 and 3.6.1.4	4.5.7
High temperature	3.2.1, 3.3.2.2 and 3.6.1.4	4.5.8
Salt fog	3.3.4	4.5.9
Vibration	3.3.5	4.5.10
Shock	3.3.6	4.5.11
Fungus	3.3.7	4.5.12
Humidity	3.3.8	4.5.13

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4.4.6 Quality conformance test report (type III, style B only). When specified (see 6.1), a quality conformance inspection report shall be furnished for group A tests. Quality conformance inspection reports in accordance with paragraph 6.5 shall be furnished for group B and C tests.

4.5 Test procedures.

4.5.1 Surface examination. Each sample psychrometer shall be visually and dimensionally examined to verify conformance with the requirements of this specification. Any psychrometer containing one or more of the following defects shall be rejected:

TABLE III. Classification of defects

Categories	Defects
Critical:	None defined.
Major:	
101	Types of psychrometers not as specified (see 1.2.1).
102	Materials defective, or not in strict accordance with specifications (see 3.2 and 3.2.1).
103	Corrodible materials not properly protected from salt air or salt spray (see 3.2.1).
104	Improper finishes used with evidence of cracking and peeling (see 3.2.1).
105	Nonanodized aluminum parts used in contact with nonferrous items (see 3.2.1).
106	One each wet-bulb and dry-bulb thermometer not supplied with each psychrometer as specified (see 3.3.1).
107	Thermometers not in accordance with specifications as follows: (a) Stem not as specified (see 3.3.2.1). (b) Bore of stem not round or oval (see 3.3.2.1). (c) Strip of opalescent glass not incorporated in stem (see 3.3.2.1). (d) Expansion space in tube not provided as specified (see 3.3.2.2 and 3.3.2.2.1). (e) Configuration of bulb not as specified, nor concentric with tube (see 3.3.2.3). (f) Evidence of foreign matter in bulb (see 3.3.2.3). (g) Physical dimensions of bulb not within specified limits (see 3.3.2.3). (h) Provisions for removal of wick for wet-bulb type not provided as specified (see 3.3.2.3). (i) Indicating mercury dull and dirty (see 3.3.2.4). (j) Terminal nib fused to tube, not of proper size or shape as specified (see 3.2.2.4.2). (k) Length of graduated scale not as specified (see 3.3.2.5). (l) Graduation lines on scale not clear, straight, perpendicular to axis, uniformly spaced, of proper length or numbered as specified (see 3.3.2.5). (m) Range of thermometers not within minimum limits as specified (see 3.3.2.5.1). (n) Total length of bulb and stem not as specified (see 3.3.2.6). (o) Wicks for wet bulb thermometer not of specified diameter or length (see 3.3.11). (p) Marking pigment not of proper color or readability not satisfactory (see 3.3.2.9 and 3.3.2.10). (q) Thermometer markings not as specified (see 3.3.2.8). (r) Housing not as specified (see 3.6.3).
108	Wall mounted (type I) psychrometers not in accordance with specifications as follows (see 3.4). (a) Design and material for back not as specified (see 3.4.1). (b) Means for easily removing and remounting thermometer with back not provided as specified (see 3.4.1). (c) Stem of thermometer not properly secured to metal back as specified (see 3.4.1.1).

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TABLE III. Classification of defects (Con.)

Categories	Defects
	(d) Brackets for securing backs to mounting board not as specified (see 3.4.2).
	(e) Mounting board not to proper dimensions, rough surface, knotty and not provided with means for hanging as specified (see 3.4.3).
	(f) Thumb nuts for securing metal back to bracket not provided as specified (see 3.4.3.1).
	(g) Water reservoir not within specified dimension limits (see 3.4.4).
	(h) Cap of reservoir not fitted properly as specified (see 3.4.4).
	(i) Spout for reservoir not of proper shape, size, or located as specified (see 3.4.4.1).
	(j) Reservoir mounting on mounting board not as specified (3.4.4.2).
	(k) Physical dimensions and design of shelter not as specified (see 3.4.5).
109 ----------	Sling type (type II) psychrometers not in accordance with specifications as follows (see 3.5):
	(a) Thermometer frame configuration and dimensions not in accordance with specification (see 3.5.1).
	(b) Thermometer frame not straight and true in area for mounting thermometers or evidence of stain on thermometers and rough finished with burrs and sharp edges (see 3.5.1.1).
	(c) Holes for terminal nibs, securing clips and seivel hooks not located as specified or drilled to required dimensions (see 3.5.1).
	(d) Means for moistening wick of wet-bulb thermometer not provided when specified (see 3.5.2).
	(e) Handle and sling not as specified. Bearing not provided as specified, or inoperative, and swivel hook configuration and length not as required (see 3.5.3)
110 ----------	Fan ventilated, battery operated (type III), psychrometer not in accordance with specifications as follows (see 3.6).
	(a) Overall dimensions and weight not within specified limits (see 3.6.1).
	(b) Supports in housing not rugged enough to withstand rough handling (see 3.6.3.1).
	(c) Proper access to permit replacement of batteries not provided (see 3.6.3.2).
	(d) Means for coupling fan and motor not provided as specified (see 3.6.4).
	(e) Path of air stream not directed as specified (see 3.6.4).
	(f) Proper access to permit removal and replacement of fan motor unit not provided (see 3.6.4.2).
	(g) Positive means for starting and stopping motor not provided as specified (see 3.6.4.3).
	(h) Thermal shields for thermometers not provided or designed as specified (see 3.6.5).
	(i) Illumination for thermometer scales not provided as specified (see 3.6.6).
	(j) Controls to effect independent operation of motor and illumination not provided (see 3.6.4.3).
	(k) Specified method of mounting thermometers not provided (see 3.6.7).
	(l) Water bottle and stowage not provided as specified (see 3.6.8)
	(m) Battery requirements not as specified (see 3.6.3.2).
	(n) External sources of heat influence readings (see 3.6.1.2).

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TABLE III. Classification of defects (Con.)

Categories	Defects
111 ----------	Fan ventilated, 115 volt, a.c. and d.c. (type IV), psychrometer not in accordance with specifications as follows (see 3.9).
	(a) Overall dimensions and weight not within specified limits (see 3.6.1).
	(b) Supports in housing not rugged enough to withstand rough handling (see 3.6.3.1).
	(c) Universal type motor, a.c. and d.c., not provided as specified (see 3.9.2).
	(d) Means for coupling fan and motor not provided as specified (see 3.6.4).
	(e) Path of air stream not directed as specified (see 3.6.4).
	(f) Proper access to permit removal and replacement of fan motor unit not provided (see 3.6.4.2).
	(g) Positive means for starting and stopping motor not provided as specified (see 3.6.3).
	(h) Thermal shields for thermometers not provided or designed as specified (see 3.6.5).
	(i) Illumination for thermometer scales not provided as specified (see 3.6.6).
	(j) Controls to effect independent operation of motor and illumination not provided (see 3.6.4.3).
	(k) Size, type, and connections for power cord not as specified (see 3.9.3).
	(l) Specified method of mounting thermometers not provided (see 3.6.7).
	(m) Water bottle and stowage not provided as specified (see 3.6.8).
	(n) External sources of heat influence readings (see 3.6.1.2).
112 ----------	Sling, pocket-type (type V) psychrometers not in accordance with specifications as follows (see 3.10):
	(a) Thermometer frame configuration not in accordance with specification (see 3.10.1).
	(b) Thermometer frame not straight and true in area for mounting thermometers (see 3.10.1.1).
	(c) Thermometer frame not straight or true with evidence of strain on thermometers and rough finished with burrs and sharp edges (see 3.10.1.1).
	(d) Punch outs for button-tops and holes for securing clips not located as specified (see 3.10.1).
	(e) Means for moistening wick of wet-bulb thermometer not provided when specified (see 3.10.2).
	(f) Handle and sling not as specified (see 3.10.3).
Minor:	
201 ----------	Psychrometric tables not supplied as specified.
202 ----------	Thermometer frames not straight or true with evidence of strain on thermometers and rough finished with burrs and sharp edges.
203 ----------	Instrument case not furnished in accordance with specifications.
204 ----------	Identification where required not as specified.
205 ----------	Mounting brackets for type III and type IV psychrometers not as specified.
206 ----------	Repair parts and supplemental equipment not as specified or missing.
207 ----------	Markings, packaging, and preservation not as specified.
208 ----------	Technical manual not supplied as specified.

4.5.2 Thermometer accuracy. The inspector shall obtain from the contractor two thermometers of which the calibration of each is known to be within 0.1°F over the required test range of the thermometers to be tested, and shall use these thermometers as the master in making the test for accuracy of the psychrometer thermometers.

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4.5.2.1 The thermometers to be tested shall be immersed to the top of the liquid column in a liquid bath which shall be constantly agitated in order that all instruments may be exposed to the same temperature. The master thermometers shall be immersed to the top of the liquid column in the same bath as the thermometers being checked, and shall be centrally located with respect to them. Both the master and test thermometers shall have their bulbs placed in approximately the same level in the bath. The temperature of the liquid bath shall be held constant to within 0.1°F of the test point selected (see 4.5.2.3). Data shall not be taken until both the master and test thermometers have responded completely to the temperature of the bath. When test data are taken, the line of sight from the observer's eye shall be normal to the stem of the instrument.

4.5.2.2 The contractor shall furnish the Government inspector a statement identifying the check points which were used in graduating the thermometer tube. In the test for accuracy, all test temperatures shall be selected at points intermediate between these points, unless a check point is also an ice point (see 4.5.2.3).

4.5.2.3 The thermometers to be tested shall be compared with the master thermometer in an ice bath and at a series of points taken over the entire range. Test points shall be selected such that the uppermost and lowermost points shall fall within 0.5°F of the end points of the specified ranges, and the interval between consecutive test points shall not be closer than 10 scale divisions and no further apart than every 50 scale divisions. An ice point should always be included as one of the test points, where an ice point is covered by the range of the thermometer. Accuracy tests shall be made with the indicator rising and also with the indicator falling with respect to each test point over the range of the thermometer. The thermometer shall indicate true temperature within the tolerance of 3.3.2.7 in both directions.

4.5.3 Expansion space (except type V). The thermometers shall be immersed up to the shoulder of the top of the bulb chamber for 5 minutes in a liquid bath having a temperature of 50°F in excess of its maximum scale reading or to the maximum design nonoperating temperature (see 3.3.3), whichever is greater. The bulb shall then be allowed to cool in air to ambient temperature. The thermometer scale and glass tube shall not be clouded by volatilization of the heat transfer material around the glass bulb, at any temperature indicated on the scale. After this test, the thermometers shall indicate true temperatures within the tolerances of 3.3.2.7 in both direction.

4.5.3.1 Expansion space (type V only). The thermometer shall be immersed completely for five minutes in a water bath having a temperature of 130°F. The thermometer shall then be cooled in air to ambient temperature. After this test, the thermometer shall indicate true temperatures within the tolerances of 3.2.2.8 in both directions.

4.5.4 Color stability. The organic liquid column shall be free from distillation and shall still be easily readable after being subjected to a sunshine test in accordance with method 505, procedure II, of MIL-STD-810.

4.5.5 Test on handles and slings (types II and V only). Handles and slings for types II and V psychrometers shall be tested as specified in 4.5.5.1 and 4.5.5.2.

4.5.5.1 Type II. A 3-pound weight shall be attached to the sling in place of a psychrometer. The weight shall be rotated about the handle at a rate of approximately 3 revolutions per second for 10 seconds. There shall be no damage to the handle and sling.

4.5.5.2 Type V. A 1-pound weight shall be attached to the sling in place of a psychrometer. The weight shall be rotated about the handle at a rate of approximately 3 revolutions per second for 10 seconds. There shall be no damage to the handle and sling.

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4.5.6 Air velocity (types III and IV). The air speed over the wet- and dry-bulb thermometers of the sample psychrometer shall be determined by operating the psychrometer with a hot wire anemometer, Kafa-thermometer, or other approved temperature air speed device substituted in place of the thermometer bulbs, to determine compliance with 3.6.4.4. To avoid errors in making the air speed measurement, the air speed probe employed shall occupy the same space and be of the approximate same shape and size as the thermometer bulb replaced.

4.5.7 Low temperature (type III, style B). The low temperature test shall be conducted in accordance with MIL-E-16400 except batteries shall not be installed while the psychrometer is subjected to storage temperature.

4.5.8 High temperature (type III, style B). The high temperature test shall be conducted accordance with MIL-E-16400 except batteries shall not be installed while the psychrometer is subjected to storage temperature.

4.5.9 Salt fog (type III, style B). A salt fog test shall be conducted in accordance with MIL-E-16400 for exposed equipment.

4.5.10 Vibration test (type III, style B). The psychrometers in their cases with batteries and accessories in place shall be tested in accordance with type I of MIL-STD-167-1.

4.5.11 Shock test (type III, style B). The psychrometers in their cases with batteries and accessories in place shall be subjected to grade A class II, type A, lightweight shock test in accordance with MIL-S-901 except the 5-foot hammer blows may be omitted. The psychrometer shall be fully functional at the conclusion of the test.

4.5.12 Fungus test (type III, style B). The psychrometers shall be subjected to the fungus test specified in method 508 of MIL-STD-810. This test shall be conducted after all other applicable tests have been completed.

4.5.13 Humidity test (type III, style B). The psychrometers shall be subjected to the humidity test specified in MIL-E-16400. Batteries shall be in place during the test. Performance measurements required by method 507 of MIL-STD-810 shall consist of determination of the temperature and humidity conditions in the test chamber by the psychrometers under test and comparison of the results with actual humidity and temperature conditions established independently during the test cycle. Thermometer accuracy shall conform to 3.3.2.7 and the calculator of 3.3.10.1 shall indicate the corresponding relative humidity within the accuracy requirements of 3.6.10.1. Should any particular test condition be off the scale of the thermometers on the psychrometers under test, operation of the fan and light(s) will be considered satisfactory performance for that test point only.

4.5.14 Pigment stability. All etched graduations and numerals shall be examined to determine compliance with 3.3.2.9.

4.5.15 Stain durability. Thermometers having permanent stained scales imprinted in the glass (see 3.3.2.5) shall have either a portion or all of the graduated scale exposed to a 20 percent caustic soda solution at approximately 65°C (149°F) for one-half hour, followed by a similar exposure to 50 percent sulphuric acid at approximately 65°C (149°F) for one-half hour. This cycle shall be repeated twice, for a total of three cycles. There shall be no discernable deterioration of coloring on the test thermometers when compared to other untreated thermometers.

4.6 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of section 5 of this specification. Defects shall be scored in accordance with table II. For examination of interior packaging, the sample unit shall be one shipping container identical to each item being packed, fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

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TABLE IV. Classification of preparation for delivery defects

Examine	Defects
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Material	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, or distortion of container.
Contents	Gross weight exceeds the requirements of the box specification.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2 (see 6.2).)

5.1 Preservation and packaging. Preservation (when applicable) and packaging shall be level A or Commercial as specified (see 6.1).

5.1.1 Level A. Psychrometers and accessories shall be individually preserved and packaged in accordance with MIL-P-116 as follows: Types I, II, IIIA, IV, V by method III, and type IIIB by method II.

5.1.2 Commercial. The psychrometer with accessories shall be packaged in unit containers normally used in commercial practice.

5.2 Packing. Packing shall be level A, or Commercial as specified (see 6.1).

5.2.1 Level A. Types I, II, IIIA, IV and V psychrometers and accessories packaged as specified in 5.1.1 shall be packed in close-fitting boxes conforming to PPP-B-636, class weather resistant or to PPP-B-640, class 2. Close, waterproof seal and strap the boxes in accordance with the appendix to the applicable box specification. Type IIIB psychrometers and accessories, packaged in accordance with 5.1.1, shall be packed in close-fitting overseas type boxes conforming to PPP-B-591, PPP-B-621, PPP-B-585, PPP-B-636 or to PPP-B-601, at the option of the contractor. Shipping containers shall have case liners conforming to MIL-L-10547, and shall be closed and sealed in accordance with the appendix thereto. Case liners for boxes conforming to PPP-B-636 may be omitted, provided all joints of the boxes are sealed with tape as specified in the appendix of the box specification. Boxes shall be closed and reinforced as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.2 Commercial. The psychrometers and accessories shall be packed in shipping containers to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

5.3 Unitization. When shipments to Government depots are full car or truckload, the shipping containers shall be unitized for shipment and handling in accordance with normal commercial practice. The unitized load shall not exceed 2,500 lbs. in weight, 63 inches in height, 56 inches in length, and 45 inches in width.

5.4 Marking. Marking shall be in accordance with 5.4.1 or 5.4.2 as specified (see 6.1).

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5.4.1 Civil agencies. Packages, shipping containers and unitized loads (when applicable) shall be marked in accordance with Fed. Std. No. 123.

5.4.2 Military agencies. Packages, shipping containers and unitized loads (when applicable) shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2.1).
(c) Whether brass, stainless steel, or aluminum material is required (see 3.2.1).
(d) Specify when thermometer indicator is to be optional organic liquid (see 3.2.2.4.1).
(e) When mounting is not required on the thermometers (see 3.3.2).
(f) Length of graduated scale if other than specified (see 3.3.2.5).
(g) Thermometer range (see 3.3.2.5.1).
(h) Name or initials and serial number for marking, if desired (see 3.3.2.8).
(i) Fastening for wick if other than specified (see 3.3.2.11).
(j) Whether shelter is required for type I psychrometer (see 3.4.5).
(k) When water reservoir is required for types II or V (see 3.5.2 or 3.10.2).
(l) When handle and sling is not required for type II (see 3.5.3).
(m) When protective holder is required for type II (see 3.5.4).
(n) When an instrument container is not required for types II or V (see 3.5.5 or 3.10.5).
(o) When psychrometric table is not to be furnished (see 3.5.6).
(p) Whether batteries are in accordance with Federal or Military Specification (see 3.6.4.1).
(q) Stock number and serial numbers for name plate (see 3.6.9).
(r) Whether carrying case for types III and IV will provide for storage of repair parts (see 3.6.10.1 and 3.9.1).
(s) Whether a bracket for mounting types III and IV required (see 3.6.11 and 3.9.1).
(t) Repair parts and quantities required (see 3.7).
(u) Whether group C tests are required (see 4.4.5).
(v) Whether inspection reports are required for group A tests (see 4.4.6).
(w) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).
(x) Whether civil or military marking is required (see 5.4).

6.2 Contract data requirements. When this specification is used in a procurement invoking the provisions of Armed Services Procurement Regulations (ASPR), paragraph 7-104.9(n) and which incorporates a DD Form 1423 Contract Data Requirements List (CDRL) the data requirements identified below will be developed as specified in the cited Data Item Description (DID) and delivered in accordance with such CDRL. When the ASPR provisions are not invoked, the data specified below shall be delivered in accordance with the contract requirements.

Spec. paragraph	Data requirement	Service	Applicable DID	Options
3.5	Technical manual standard basic issue	SH	-	Commercial per MIL-M-15071
4.3.1	First article inspection report	SH	UDI-T-23473	
4.4.6	Quality conformance inspection report	SH	UDI-T-23473	

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Spec. paragraph	Data requirement	Service	Applicable DID	Options
4.5.9	Vibration test report	SH	UDI-T-23762	
4.5.10	Shock test procedures	SH	UDI-T-23753	
	Shock test report		UDI-T-23753	
	Procedures, post shock test functional testing		UDI-T-23755	

(Copies of data item descriptions (DIDs) required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply).

6.3 Data submission. For Navy procurement, the CDRL should specify that test reports shall be submitted to the Naval Ship Engineering Center, Mechanicsburg Division, Mechanicsburg, PA 17055 for approval not later than 30 days after the applicable test(s).

MILITARY INTERESTS:

Army - Sig.
Navy - Sh.

Preparing activity:

GSA-FSS

Project No. 6685-0353

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 50 cents each.

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